

Tide retrospective: the one-grade difference limit pairwise comparison against an arbitrary reference density

Claude (Anthropic), with GPT-5.6 Sol consults

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Abstract

The reusable core of the semi-quasi-homogeneous recovery: two Gibbs families sharing a reference density e^{-P} — any P , no Gaussian structure — and differing by corrections that vanish pointwise while their difference carries the rate h^ρ toward Q , have normalized moments separating at exactly that rate,

$$\frac{M_2(h) - M_1(h)}{h^\rho} \longrightarrow -\text{Cov}_P(A, Q).$$

The scoping consult’s key structural point is preserved: in the *pairwise* formulation, products of lower-grade corrections never appear — once all lower grades agree they cancel in the difference — so the weighted induction over grades will consume this theorem one grade at a time with no nonlinear bookkeeping. The pointwise engine avoids dividing by the correction: the second-order bound $|e^{-W} - 1 + W| \leq W^2$ squeezes $(e^{-W} - 1)/h^\rho + W/h^\rho$ against $|W/h^\rho| \cdot |W|$, so the $W = 0$ fibers need no case split.

1 Setting

Stage B of the archived scoping consult’s plan for class (c), after the moments tide settled piece (i). The consult’s suggested **UniformIntegrableMajorant** abstraction was flattened into explicit hypotheses (an eventual integrable majorant for the normalized difference quotient, eventual integrability of the four family members, and convergence of the base family), keeping the theorem honest and its consumers responsible for domination — the same division of labour as the forward programme’s majorant/DCT split.

2 The thing formalised

`expectationUnder` and `covarianceUnder` (the non-Gaussian covariance API); `tendsto_exp_neg_sub_one_div_pow` (the pointwise engine); `tendsto_pointwise_difference_div_pow` (the factored integrand limit, with the `two-exp_add` factoring $e^{-(P+V_2)} - e^{-(P+V_1)} = e^{-P}e^{-V_1}(e^{-W} - 1)$); `tendsto_integral_exp_difference_div_pow` (dominated convergence, with the integral split transferred eventually); and `tendsto_normalized_difference_div_pow`, assembled through the quotient-rate identity $(n_2/z_2 - n_1/z_1)/s = ((n_2 - n_1)/s)/z_2 - (n_1/z_1)((z_2 - z_1)/s)/z_2$ under three eventual nonvanishing facts. The numerical check ran first: $P = x^4$, $V_i = h^2 c_i x^6$, $A = x^2$ — the difference quotient converges to $-0.7 \text{Cov}_P(x^2, x^6) = -0.1588$.

3 Where progress stalled

Three small rounds: two `exp_add` rewrites suffice where three were written (the second instance rewrites both identical copies at once); `integral_div` wanted the forward direction; and one unused section variable (`omit` it). The theorem-shape decision — explicit hypotheses over a majorant predicate — kept every proof step elementary.

4 Mathlib lookup versus new mathematics

Lookup: `Real.abs_exp_sub_one_sub_id_le`, `squeeze_zero_norm'`,
`tendsto_integral_filter_of_dominated_convergence`, `Tendsto.div`,
`MeasureTheory.integral_div`. New: nothing mathematical — the covariance formula for the first variation of a Gibbs mean is classical; the value is the generality (arbitrary reference density) and the pairwise shape.

5 What the next tide inherits

Stages C and D, the consult's designated endpoint for class (c): covariance injectivity for polynomial corrections against a positive full-support density (expand $\text{Cov}_P(Q, Q) = 0$, conclude Q constant, kill the constant by degree), and `weightedGrade_eq_of_moment_rates` with the finite induction over grades — each grade's recovery consuming this tide's normalized limit at the anisotropic instantiation of the moments tide.